

MCQs Answers (1–15)

1. Main function of a ticketing tool?

- A. Data storage
- ☒ B. Issue tracking
- C. Network routing
- D. File sharing

Explanation:

Ticketing tool is used to **log, track, and manage user issues** in IT support.

2. Brain of the computer?

- A. RAM
- ☒ B. CPU
- C. HDD
- D. PSU

Explanation:

CPU performs all **processing and decision making**, so it is called brain.

3. Valid IPv4 address?

- A. 300.1.1.1
- ☒ B. 192.168.1.1
- C. 256.256.256.256
- D. 192.168.1

Explanation:

IPv4 range is **0–255**, so only option B is valid.

4. Default subnet mask of Class C?

- A. 255.0.0.0
 - B. 255.255.0.0
 - ☒ C. 255.255.255.0
 - D. 255.255.255.255
-

5. Protocol using Dijkstra algorithm?

- A. EIGRP
- B. RIP
- ☒ C. OSPF
- D. BGP

Explanation:

OSPF uses **Shortest Path First (Dijkstra algorithm)**.

6. Default Administrative Distance of OSPF?

- A. 90
 - B. 100
 - ☒ C. 110
 - D. 120
-

7. ACL type filtering source & destination IP?

- A. Standard ACL
 - ☒ B. Extended ACL
 - C. Dynamic ACL
 - D. Reflexive ACL
-

8. Default VLAN on Cisco switch?

- A. VLAN 0
- ☒ B. VLAN 1
- C. VLAN 100
- D. VLAN 10

9. Protocol preventing switching loops?

- A. ARP
- ☒ **B. STP**
- C. DHCP
- D. FTP

Explanation:

STP (Spanning Tree Protocol) **prevents loops in switching network.**

10. Default HSRP priority?

- A. 90
 - ☒ **B. 100**
 - C. 110
 - D. 120
-

11. Purpose of trunk port?

- A. Connect end devices
 - ☒ **B. Carry multiple VLANs**
 - C. Assign IP addresses
 - D. Filter traffic
-

12. Command to bundle interfaces in Port Channel?

- A. ip route
- ☒ **B. channel-group**
- C. switchport mode
- D. vlan database

Explanation:

Used to create **EtherChannel / Port Channel.**

13. What is ROAS used for?

☒ **A. Routing between VLANs**

B. Assigning IP addresses

C. Loop prevention

D. Port security

Explanation:

ROAS = Router on a Stick → used for **inter-VLAN routing**.

14. What does CIDR stand for?

☒ **A. Classless Inter-Domain Routing**

B. Common Internet Data Routing

C. Central IP Domain Routing

D. Classful Internet Data Routing

15. Default action in port security violation?

A. Port restarts

☒ **B. Port shuts down**

C. Traffic allowed

D. VLAN deleted

Explanation:

Default mode is **shutdown** (interface goes into err-disabled state).

Ticketing Tool Questions (1–4)

1. What is a Ticketing Tool?

Definition (What it is)

A **Ticketing Tool** is a software system used to **track, manage, and resolve IT issues or service requests** .

 Each issue is recorded as a **“ticket”**.

Purpose (Why it is used)

- Organize and track **user issues**
 - Assign tasks to **IT teams**
 - Monitor progress and resolution
 - Maintain **history of problems**
-

How it Works

1. User reports an issue
 2. Ticket is created
 3. Assigned to support engineer
 4. Issue is worked on
 5. Ticket is resolved and closed
-

Real-Life Example

Like a **complaint system in a company** .

- You report a problem
- It gets logged
- Assigned to someone

- Fixed and closed
-

Important Points

- Common tools: ServiceNow, Jira, Zendesk
 - Helps in **tracking & accountability**
-

2. What is SLA in a Ticketing System?

Definition

SLA (Service Level Agreement) is the **time limit within which a ticket must be responded to or resolved** 🕒.

Purpose

- Ensure **timely resolution**
 - Improve **customer satisfaction** 😊
 - Set **clear expectations**
-

How it Works

- Ticket is created
 - SLA timer starts 🕒
 - Must:
 - Respond within X time
 - Resolve within Y time
-

Real-Life Example

Like a **delivery promise**  :

- “Your order will arrive in 2 days”
 That’s an SLA
-

Important Points

- Response SLA
 - Resolution SLA
 - Breach = SLA violation 
-

3. What is Ticket Escalation?

Definition

Ticket Escalation is the process of **forwarding a ticket to a higher-level support team** when it cannot be resolved at the current level  .

Purpose

- Handle **complex issues**
 - Ensure **quick resolution**
 - Involve **experts when needed**
-

How it Works

1. Issue is assigned to Level 1
2. If not resolved → escalated
3. Sent to:
 - Level 2 or Level 3 team

4. Experts fix the issue

Real-Life Example

Like asking a **senior engineer**  when a junior cannot fix something

Important Points

- Types:
 - **Functional escalation** (to higher team)
 - **Hierarchical escalation** (to manager)
-

4. Which Field in a Ticket is Used to Prioritize Issues?

Answer

 The field used is **Priority** 

Purpose

- Decide **which issue to solve first**
 - Manage **critical vs normal issues**
-

How it Works

Priority is usually based on:

- **Impact** (how many users affected)
 - **Urgency** (how quickly it needs fixing)
-

Common Priority Levels

Priority Meaning

P1  Critical (system down)

P2  High

P3  Medium

P4  Low

Real-Life Example

Like a **hospital emergency system**  :

- Critical patient → treated first
 - Minor issue → later
-

Important Points

- Priority = Impact + Urgency
 - Helps meet **SLA targets**
-

Quick Summary

- Ticketing tool = **Issue tracking system**
 - SLA = **Time commitment**
 - Escalation = **Move to higher support**
 - Priority = **Decides urgency**
-



Desktop Hardware Questions (1–5)



1. What is RAM?



Definition (What it is)

RAM (Random Access Memory) is a type of **temporary memory** used by a computer to store data that is currently in use ⚡.



It is **volatile memory** (data is lost when power is off)



Purpose (Why it is used)

- Stores **running programs and data**
 - Helps CPU access data **quickly**
 - Improves **system speed and performance**
-



How it Works

1. You open an application
 2. Data loads from hard disk → RAM
 3. CPU processes data from RAM
 4. When system shuts down → data is cleared
-



Real-Life Example

RAM is like a **working table** 🧑💻:

- Bigger table → more work at once
 - Small table → slow work
-

Important Points

- Faster than storage
 - Measured in GB (8GB, 16GB)
 - Types: DDR3, DDR4, DDR5
-

2. What is ROM?

Definition

ROM (Read Only Memory) is **permanent memory** that stores important system instructions .

 It is **non-volatile** (data stays even when power is off)

Purpose

- Stores **boot instructions (BIOS/UEFI)**
 - Helps system **start (boot)**
-

How it Works

1. Power ON system
 2. ROM runs firmware
 3. Loads OS from storage
-

Real-Life Example

ROM is like a **book**  :

- Content is fixed
 - You read it, but don't change it often
-

Important Points

- Non-volatile
 - Stores firmware
 - Slower than RAM
-

3. What is the Function of a Motherboard?

Definition

The **motherboard** is the **main circuit board** that connects all components of a computer .

Purpose

- Connects CPU, RAM, storage, etc.
 - Allows components to **communicate**
 - Distributes **power**
-

How it Works

- CPU processes data
 - RAM stores temporary data
 - Motherboard connects everything via circuits
-

Real-Life Example

Motherboard is like a **city road system** .

- Connects all buildings (components)
 - Enables movement (data flow)
-

Important Points

- Backbone of computer
 - Contains slots, ports, chipset
-

4. What is SMPS?

Definition

SMPS (Switched Mode Power Supply) is a device that **converts AC power into DC power** for computer components ⚡.

Purpose

- Supplies **power to all components**
 - Ensures correct voltage levels
-

How it Works

1. Takes AC power from wall
 2. Converts to DC
 3. Distributes to:
 - Motherboard
 - CPU
 - Hard disk
-

Real-Life Example

SMPS is like a **power adapter** 🔌 :

- Converts electricity into usable form

Important Points

- Provides stable power
 - Measured in watts (e.g., 450W, 600W)
-

5. List Motherboard Components and Their Role

Main Components & Roles

Component	Role
CPU Socket 	Holds the processor
RAM Slots	Holds memory modules
Chipset	Controls communication
BIOS/UEFI	Starts the system
Power Connector 	Supplies power
Expansion Slots (PCIe)	Add GPU, NIC etc.
Storage Connectors (SATA/M.2)	Connect HDD/SSD
I/O Ports	USB, HDMI, LAN
CMOS Battery 	Saves BIOS settings

Purpose

- Each component helps system **function properly**
-

Real-Life Example

Like parts of a car 🚗 :

- Engine = CPU
 - Fuel = Power
 - Dashboard = BIOS
-

Important Points

- All components work together
 - Fault in one can affect entire system
-

Quick Summary

- RAM = Temporary memory
 - ROM = Permanent memory
 - Motherboard = Main connection board
 - SMPS = Power supplier
 - Components = Work together for system operation
-

IP Addressing Questions (1–5)

1. What is an IP Address?

Definition (What it is)

An **IP Address (Internet Protocol Address)** is a **unique identifier assigned to each device on a network** 📶.

👉 It helps devices **find and communicate with each other**

Purpose (Why it is used)

- Identify devices on a network
 - Enable **communication between devices**
 - Help in **routing data across networks**
-

How it Works

1. Device gets an IP address
 2. Sends data to another IP
 3. Network uses this IP to deliver data
-

Real-Life Example

Like a **home address** 🏠:

- Helps courier deliver packages
👉 Without it, delivery is impossible
-

Important Points

- Two types: IPv4 & IPv6
- Works at **Layer 3 (Network Layer)**

2. What is a Public IP Address?

Definition

A **Public IP Address** is an IP address that is **accessible over the Internet** .

 Assigned by ISP (Internet Service Provider)

Purpose

- Allows communication over the **Internet**
 - Identifies your network globally
-

How it Works

- Router gets public IP
 - Used to communicate with external networks
-

Real-Life Example

Like your **house address on Google Maps** .

- Anyone can find you using it
-

Important Points

- Globally unique
 - Limited in number
 - Used with NAT
-

3. What is a Private IP Address?

Definition

A **Private IP Address** is used **within a local network** and is **not accessible on the Internet**  .

Purpose

- Internal communication
 - Saves public IPs
 - Improves security
-

How it Works

- Devices get private IP
 - Router converts it using NAT
-

Real-Life Example

Like **room numbers in a hotel**  :

- Used inside only
-

Important Ranges

- 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255
-

4. What is IPv6?

Definition

IPv6 (Internet Protocol Version 6) is the **new version of IP addressing** designed to replace IPv4 .

Purpose

- Solve **IP address shortage**
 - Improve **security and performance**
-

How it Works

- Uses **128-bit address**
- Written in hexadecimal

Example:

2001:0db8:85a3::8a2e:0370:7334

Real-Life Example

Like expanding from **small city to big city** .

- More space for addresses
-

Important Points

- Much larger address space
 - Built-in security (IPSec)
 - No need for NAT
-

5. What is APIPA?

Definition

APIPA (Automatic Private IP Addressing) is when a device **assigns itself an IP address** if DHCP fails 📶 .

Purpose

- Provide **temporary local communication**
 - Help in **troubleshooting DHCP issues**
-

How it Works

1. DHCP request fails
 2. Device assigns:
169.254.x.x
 3. Used only for local communication
-

Real-Life Example

Like a **temporary ID card** 🪪 :

- Works locally but not globally
-

Important Points

- Range: 169.254.0.0 – 169.254.255.255
 - No Internet access
 - Indicates DHCP issue
-

Quick Summary

- IP Address = Device identity
 - Public IP = Internet access
 - Private IP = Internal use
 - IPv6 = Future addressing
 - APIPA = Backup IP
-

Subnetting Questions (1–5)

1. What is Subnetting?

What it is

Subnetting is the process of dividing a large network into smaller, manageable networks called **subnets**.

Why it is used (Purpose)

- Improves network performance 
 - Reduces network congestion
 - Enhances security 
 - Makes IP address management easier
-

How it works

1. A large network (like 192.168.1.0/24) is taken
 2. Some bits from the **host portion** are “borrowed”
 3. This creates multiple smaller networks (subnets)
 4. Each subnet has its own range of IP addresses
-

Real-life example

Think of a **big apartment building** 🏢:

- Entire building = Network
- Each floor = Subnet
- Each apartment = Device (IP address)

Instead of everyone sharing one space, people are organized floor-wise — much more efficient!

★ Important points

- Subnetting uses **subnet masks or CIDR notation**
 - Each subnet has:
 - Network address
 - Broadcast address
 - Usable IP range
 - Helps in **routing and traffic control**
-

2. What is a Subnet Mask?

✅ What it is

A subnet mask is a number that defines **which part of an IP address is the network and which part is the host.**

Example:

- IP: 192.168.1.10
 - Subnet Mask: 255.255.255.0
-

Why it is used

- Helps devices understand:
 - “Is this device in my network or not?” 🤔
 - Required for proper communication between devices
-

How it works

1. Subnet mask is written in binary
 2. It separates:
 - Network bits (1s)
 - Host bits (0s)
 3. Device uses it to identify:
 - Network address
 - Host portion
-

Real-life example

Think of a **phone number** 📞 :

- Area code = Network
- Personal number = Host

The subnet mask tells you where the area code ends and the personal number begins.

Important points

- Common masks:
 - 255.255.255.0 (/24)
 - 255.255.0.0 (/16)
- Written in **dotted decimal** or **CIDR format**

- Essential for subnetting
-

3. What is a Broadcast Address?

✅ What it is

A broadcast address is a special IP address used to send data to **all devices in a network at once** 📡

🎯 Why it is used

- To communicate with all devices simultaneously
 - Used in protocols like:
 - ARP
 - DHCP
-

⚙️ How it works

1. All host bits are set to **1**
 2. Example:
 - Network: 192.168.1.0/24
 - Broadcast: 192.168.1.255
 3. When a message is sent to this address:
 - Every device in the subnet receives it
-

🏠 Real-life example

Like making an **announcement in a classroom** 🎤 :

- Teacher speaks once
 - Every student hears it
-

★ Important points

- Cannot be assigned to a device ❌
 - Always the **last IP** in a subnet
 - Used for **one-to-many communication**
-

4. What is a Network Address?

✅ What it is

A network address is the **first IP address of a subnet**, used to identify the network itself.

🎯 Why it is used

- Identifies the network uniquely 🌐
 - Used by routers to route traffic
-

⚙️ How it works

1. All host bits are set to **0**
 2. Example:
 - IP: 192.168.1.10/24
 - Network address: 192.168.1.0
-

🏠 Real-life example

Think of a **street name** 🏡 :

- Street name = Network
 - Houses on that street = Devices
-

★ Important points

- Cannot be assigned to devices ❌
 - Always the **first IP** in a subnet
 - Used in routing tables
-

5. What is CIDR Notation?

✅ What it is

CIDR (Classless Inter-Domain Routing) notation is a way to represent an IP address and its subnet mask using a **slash (/)**.

Example:

- 192.168.1.0/24
-

🎯 Why it is used

- Makes subnetting easier and more flexible
 - Replaces old class-based networking
 - Saves IP addresses efficiently
-

⚙️ How it works

1. The number after “/” tells how many bits are used for the network
 2. Example:
 - /24 → 24 bits for network
 - Remaining bits for hosts
 3. More bits = smaller network
-

Real-life example

Think of dividing a **pizza** 🍕 :

- Whole pizza = Network
- Number of slices = Subnet size

CIDR tells you how many slices you made.

★ Important points

- /8, /16, /24 are common
 - Smaller CIDR number → larger network
 - Larger CIDR number → smaller network
 - Used in modern networking and routing
-

🧠 Quick Summary Table

Concept	Key Idea
Subnetting	Dividing network into smaller parts
Subnet Mask	Defines network vs host portion
Broadcast Address	Sends data to all devices
Network Address	Identifies the subnet
CIDR Notation	Compact way to write subnet mask



Routing (EIGRP & OSPF) Questions (1–6)

1. What is EIGRP?



What it is

EIGRP (Enhanced Interior Gateway Routing Protocol) is a **dynamic routing protocol** used by routers to automatically share routing information.



Why it is used

- Automatically finds the best path to a destination
 - Faster than older protocols like RIP 🚀
 - Reduces manual configuration
-



How it works

1. Routers exchange routing information with neighbors
 2. Uses an algorithm called **DUAL (Diffusing Update Algorithm)**
 3. Calculates the best path based on:
 - Bandwidth
 - Delay
 4. Stores:
 - Best route (Successor)
 - Backup route (Feasible Successor)
-



Real-life example

Think of **Google Maps** 🗺️ :

- It finds the fastest route
- Also keeps a backup route in case of traffic

That's exactly how EIGRP works!

★ Important points

- Hybrid protocol (combines distance vector + link-state features)
 - Very fast convergence ⚡
 - Supports unequal load balancing
 - Mainly used in Cisco networks
-

2. What is OSPF?

✅ What it is

OSPF (Open Shortest Path First) is a **link-state routing protocol** used to find the shortest path in a network.

🎯 Why it is used

- Works well in large networks 🌐
 - Faster and more scalable than RIP
 - Vendor-neutral (not limited to Cisco)
-

⚙️ How it works

1. Routers share **link-state information**
 2. Each router builds a full **topology map** of the network
 3. Uses **Dijkstra's algorithm** to calculate shortest path
 4. Chooses path based on **cost (bandwidth)**
-

Real-life example

Think of a **city map** :

- You can see all roads
- You choose the shortest route yourself

OSPF routers do the same!

Important points

- Uses **cost metric** (lower is better)
 - Supports hierarchical design (areas)
 - Faster convergence than many protocols
 - Uses LSAs (Link-State Advertisements)
-

3. What is an OSPF Area?

What it is

An OSPF area is a **logical group of routers** used to divide a large OSPF network into smaller parts.

Why it is used

- Improves performance 
 - Reduces routing overhead
 - Makes network scalable
-

How it works

1. Network is divided into areas
2. Each area has its own topology database
3. All areas connect to **Area 0 (Backbone Area)**

4. Routers summarize routes between areas

Real-life example

Think of a **country divided into states**  :

- Each state manages its own roads
 - Highways connect states together
-

★ Important points

- Area 0 is mandatory (Backbone)
 - Types:
 - Backbone Area
 - Stub Area
 - NSSA
 - Reduces CPU and memory usage
-

4. What is a DR (Designated Router)?

What it is

A **Designated Router (DR)** is a router elected in OSPF networks to **reduce unnecessary communication**.

Why it is used

- Prevents too many routers from talking to each other
 - Reduces network traffic 
-

How it works

1. In multi-access networks (like LAN), routers elect:
 - DR (main)
 - BDR (backup)
 2. All routers send updates only to DR
 3. DR distributes updates to others
-

Real-life example

Think of a **class monitor** 🎓 :

- Students talk to the monitor
 - Monitor communicates with the teacher
-

Important points

- Election based on:
 - Priority
 - Router ID
 - Exists in broadcast networks
 - BDR takes over if DR fails
-

5. What is Convergence in Routing?

What it is

Convergence is the time it takes for all routers in a network to **learn and agree on the correct routing information**.

Why it is used

- Ensures network stability
 - Allows routers to recover after failures
-

How it works

1. Network change occurs (link down, new route)
 2. Routers exchange updates
 3. Routing tables are updated
 4. All routers reach the same view of the network
-

Real-life example

Think of **traffic updates** 🚦 :

- Road closes
 - Everyone updates their route
 - Soon all drivers know the new path
-

Important points

- Faster convergence = better network ⚡
 - EIGRP and OSPF converge quickly
 - Slow convergence can cause packet loss
-

6. What is Administrative Distance (AD)?

What it is

Administrative Distance is a value that tells a router **how trustworthy a routing source is**.

Why it is used

- Helps router choose between multiple routing protocols
-

How it works

1. Router receives routes from different protocols
 2. Each protocol has an AD value
 3. Router selects route with **lowest AD**
-

Real-life example

Think of asking directions 🙋

- Friend = somewhat reliable
- Google Maps = very reliable

You trust the most reliable source → lowest AD

Important points

Protocol	AD
Directly connected	0
Static route	1
EIGRP	90
OSPF	110
RIP	120

- Lower AD = more trusted ✓
 - Only used when multiple protocols give same destination
-

Quick Summary

Concept	Key Idea
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EIGRP	Fast hybrid routing protocol
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OSPF	Link-state, scalable protocol
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OSPF Area	Divides network for efficiency
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DR	Reduces routing traffic
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Convergence Time to update all routers

AD	Trust value of routing source
----	-------------------------------



ACL Questions (1–4)

1. What is an ACL?



What it is

An **ACL (Access Control List)** is a set of rules used in routers or switches to **permit or deny network traffic**.



Why it is used

- Controls who can access the network 
 - Improves security
 - Filters unwanted traffic
 - Controls data flow between networks
-

How it works

1. ACL contains multiple rules (called ACEs – Access Control Entries)
 2. Router checks packets **top to bottom**
 3. First match rule is applied
 4. Action is either:
 - Permit 
 - Deny 
-

Real-life example

Think of a **security guard at a gate**  :

- Checks a list
- Allows some people in
- Blocks others

That list = ACL

Important points

- Order of rules matters 
 - First match is applied (no checking further)
 - Used on router interfaces
 - Can filter:
 - IP address
 - Protocol
 - Port number
-

2. Difference between Standard and Extended ACL

✓ What it is

Two types of ACLs based on how much they can filter.

🔍 Comparison Table

Feature	Standard ACL	Extended ACL
Filters based on	Source IP only	Source + Destination + Port
Control level	Basic	Advanced
ACL Number range	1–99	100–199
Use case	Simple filtering	Detailed security

🎯 Why this difference matters

- Standard ACL = simple but limited
 - Extended ACL = more control and security 🔒
-

⚙️ How it works

- **Standard ACL**
 - Checks only who is sending traffic
 - **Extended ACL**
 - Checks:
 - Who is sending
 - Where it's going
 - What type of traffic (HTTP, FTP, etc.)
-

Real-life example

- Standard ACL = “Allow people from Mumbai” 
 - Extended ACL = “Allow people from Mumbai to enter only the office, not the server room” 
-

Important points

- Extended ACL is more commonly used
 - Standard ACL can accidentally block too much traffic
-

3. Where should Extended ACLs be placed?

What it is

Placement refers to **where in the network you apply the ACL.**

Rule (Very Important for Interviews)

 Extended ACLs should be placed as close to the source as possible

Why it is used

- Stops unwanted traffic early 
 - Saves bandwidth
 - Improves network performance
-

How it works

1. Identify source of unwanted traffic
 2. Apply ACL on router interface near that source
 3. Traffic is filtered before traveling through network
-

Real-life example

Imagine blocking spam emails :

- Better to block them at sender side
 - Instead of letting them travel across network
-

Important points

- Extended ACL → close to source
 - Standard ACL → close to destination
 - Helps reduce unnecessary traffic
-

4. What is Implicit Deny?

What it is

Implicit deny means that **any traffic not explicitly permitted is automatically denied.**

Why it is used

- Provides default security 
 - Ensures unknown traffic is blocked
-

How it works

1. ACL rules are checked top to bottom
 2. If no rule matches:
 -  Router automatically applies:
“deny all” (hidden rule)
-

Real-life example

Think of a **VIP guest list** 🚫:

- Only listed names can enter
 - If your name is not on the list → you're denied
-

★ Important points

- Implicit deny is always present ⚠️
 - You must add:
 - permit rules carefully
 - Often need a final rule like:
 - permit any (if needed)
-

🧠 Quick Summary

Concept	Key Idea
ACL	Filters network traffic
Standard ACL	Filters only source IP
Extended ACL	Filters source, destination, ports
Placement	Extended → near source
Implicit Deny	Default “deny all” at end

HSRP Questions (1–3)

1. What is HSRP?

What it is

HSRP (Hot Standby Router Protocol) is a Cisco protocol used to provide **network redundancy** by using multiple routers as a backup for each other.

Why it is used

- Prevents network failure if a router goes down ❌
 - Ensures continuous network availability ✅
 - Provides a **backup gateway** for devices
-

How it works

1. Two or more routers form an HSRP group
 2. One router becomes:
 - **Active router** (handles traffic)
 3. Another becomes:
 - **Standby router** (backup)
 4. If Active router fails:
 - Standby router takes over immediately ⚡
 5. End devices don't notice the change
-

Real-life example

Think of a **team leader and assistant**   :

- Leader does all work
- Assistant is ready to take over anytime

If leader is absent → assistant continues work without interruption

★ Important points

- Cisco proprietary protocol
 - Uses a **virtual IP and MAC address**
 - Provides **high availability (HA)**
 - Works at **Layer 3 (Network Layer)**
-

2. What is the default priority of HSRP?

✓ What it is

Priority decides **which router becomes the Active router**.

🎯 Default value

👉 **Default HSRP priority = 100**

🎯 Why it is used

- To elect the Active router
 - Higher priority = higher chance to become Active
-

⚙️ How it works

1. Routers compare priority values
 2. Router with **highest priority becomes Active**
 3. If priorities are equal:
 - Highest IP address wins
-

Real-life example

Think of a **promotion system** 🏆 :

- Higher score → leader role
 - Same score → seniority (IP address) decides
-

★ Important points

- Default = 100
 - Range: 0–255
 - You can manually increase priority
 - Preemption allows higher priority router to take back control
-

3. What is the purpose of a Virtual IP in HSRP?

✓ What it is

A **Virtual IP (VIP)** is a shared IP address used by HSRP routers as a **default gateway for devices**.

Why it is used

- Provides a **single gateway** for all devices
 - Hides the actual routers
 - Ensures seamless failover
-

⚙️ How it works

1. HSRP group creates a virtual IP
2. Devices use this IP as their gateway
3. Active router responds to this IP

4. If Active fails:
 - Standby takes over the same IP
5. Devices continue working without changes

Real-life example

Think of a **customer care number** :

- You call one number
- Behind the scenes, different agents answer

You don't know who is answering — same concept!

Important points

- Virtual IP never changes 
- Used as default gateway for hosts
- Associated with a **virtual MAC address**
- Key for **failover transparency**

Quick Summary

Concept	Key Idea
HSRP	Provides router redundancy
Default Priority	100
Virtual IP	Shared gateway for devices



Switching (VLAN, STP, Port Security, Port Channel) Questions (1–8)

1. What is a VLAN?



What it is

A **VLAN (Virtual Local Area Network)** is a logical grouping of devices in a network, even if they are physically connected to the same switch.



Why it is used

- Improves security 🔒
- Reduces broadcast traffic
- Separates departments (HR, IT, Sales)
- Makes network management easier



How it works

1. Switch ports are assigned to VLANs
2. Devices in the same VLAN can communicate
3. Devices in different VLANs need a router (or Layer 3 switch)



Real-life example

Think of an **office building** 🏢:

- Same building (switch)
 - Different departments (VLANs)
 - HR cannot directly access IT without permission
-

★ Important points

- VLAN = logical separation
 - Each VLAN is its own broadcast domain
 - VLAN IDs range: 1–4094
-

2. What is the default VLAN in most switches?

✓ Answer

👉 The default VLAN is **VLAN 1**

🎯 Why it is important

- All ports belong to VLAN 1 by default
 - Used for initial communication
-

★ Important points

- VLAN 1 cannot be deleted ✖
 - Avoid using VLAN 1 for security reasons
 - Often used for management by default
-

3. What is a Trunk Port?

✓ What it is

A **trunk port** is a switch port that carries traffic from **multiple VLANs**.

🎯 Why it is used

- Connect switches together
 - Allow multiple VLANs to pass through one link
-

How it works

1. Frames are tagged using **802.1Q tagging**
 2. VLAN ID is added to each frame
 3. Receiving switch reads the tag and forwards correctly
-

Real-life example

Think of a **highway** :

- Multiple lanes (VLANs)
 - One road (trunk link)
-

Important points

- Uses **802.1Q protocol**
 - Carries multiple VLANs
 - Native VLAN is untagged
-

4. What is STP?

What it is

STP (Spanning Tree Protocol) is used to prevent **network loops** in switched networks.

Why it is used

- Avoids broadcast storms 
 - Prevents duplicate frames
 - Ensures stable network
-

How it works

1. Switches elect a **Root Bridge**
 2. Best path to root is calculated
 3. Redundant paths are **blocked**
 4. If a link fails:
 - Blocked path becomes active
-

Real-life example

Think of **multiple roads between cities** 🚧:

- Only one is open
 - Others are closed to avoid confusion
-

★ Important points

- Prevents loops
 - Uses port states:
 - Blocking
 - Listening
 - Learning
 - Forwarding
 - Variants: RSTP (faster)
-

5. What is Port Security?

✅ What it is

Port security is a feature that **limits which devices can connect to a switch port.**

Why it is used

- Prevents unauthorized access 
 - Controls MAC addresses
 - Protects network from attacks
-

How it works

1. Configure allowed MAC addresses
 2. Set limit (e.g., 1 device per port)
 3. Switch checks incoming device MAC
 4. If violation → action is taken
-

Real-life example

Think of a **locked Wi-Fi with device limit**  :

- Only approved devices can connect
-

Important points

- Can allow:
 - Static MAC
 - Dynamic MAC
 - Supports violation modes
-

6. What happens when a security violation occurs?

What it is

A violation happens when an **unauthorized device connects**.

Actions taken (Violation Modes)

Mode	Behavior
------	----------

Protect	Drops traffic only
---------	--------------------

Restrict	Drops + logs warning
----------	----------------------

Shutdown	Disables port 
----------	---

Real-life example

Like a **security alarm**  :

- Unknown person enters
 - Alarm triggers or door locks
-

Important points

- Default mode = **Shutdown**
 - Port goes into **err-disabled state**
 - Requires manual or auto recovery
-

7. What is a Sticky MAC Address?

What it is

A sticky MAC address is a MAC address that is **automatically learned and saved by the switch**.

Why it is used

- Makes port security easier
 - No need to manually configure MACs
-

How it works

1. Device connects
 2. Switch learns MAC address
 3. Stores it in running config
 4. Treats it as allowed MAC
-

Real-life example

Think of a **smart lock** :

- Remembers your fingerprint automatically
-

Important points

- Combines dynamic + static behavior
 - Can be saved permanently
 - Useful in real networks
-

8. What are the benefits of Port Channel?

What it is

Port Channel (EtherChannel) combines **multiple physical links into one logical link**.

Why it is used

- Increases bandwidth 
 - Provides redundancy
 - Prevents STP blocking all links
-

⚙️ How it works

1. Multiple cables connect switches
2. They are grouped into one channel
3. Traffic is load-balanced across links

🏠 Real-life example

Think of a **multi-lane bridge** 🌉:

- More lanes = more traffic capacity
- If one lane fails, others still work

★ Important points

- Uses protocols:
 - LACP (standard)
 - PAgP (Cisco)
- Improves performance + reliability
- Appears as one link to STP

🧠 Quick Summary

Concept	Key Idea
VLAN	Logical network separation
VLAN 1	Default VLAN
Trunk Port	Carries multiple VLANs
STP	Prevents loops
Port Security	Controls device access
Violation	Action when unauthorized device connects

Concept	Key Idea
---------	----------

Sticky MAC	Auto-learned MAC
------------	------------------

Port Channel	Combines links for speed + redundancy
--------------	---------------------------------------



Inter-VLAN Routing / ROAS Questions (1–4)

1. What is Inter-VLAN Routing?



What it is

Inter-VLAN Routing is the process of allowing communication between different VLANs.



Why it is used

- VLANs isolate traffic (for security 🔒)
 - But sometimes departments need to communicate
 - Inter-VLAN routing enables that communication
-



How it works

1. Devices in different VLANs cannot talk directly ❌
 2. A **Layer 3 device (router or L3 switch)** is used
 3. Traffic is sent to the router
 4. Router forwards it to the destination VLAN
-

Real-life example

Think of different **departments in a company** 🏢:

- HR and IT are separate (VLANs)
 - To communicate, they go through a **manager (router)**
-

★ Important points

- Requires Layer 3 device
 - Each VLAN needs a **gateway IP**
 - Methods:
 - Legacy inter-VLAN routing
 - ROAS
 - Layer 3 switching
-

2. What is Router-on-a-Stick (ROAS)?

✓ What it is

ROAS (Router-on-a-Stick) is a method of inter-VLAN routing using **one physical router interface**.

Why it is used

- Cost-effective 💰
 - Requires fewer router interfaces
 - Common in small to medium networks
-

How it works

1. One router interface is connected to a switch
 2. That interface is divided into **subinterfaces**
 3. Each subinterface is assigned to a VLAN
 4. Trunk port carries VLAN traffic to router
 5. Router routes traffic between VLANs
-

Real-life example

Think of a **single receptionist**  handling calls for multiple departments:

- Same person
 - Different extensions (subinterfaces)
-

Important points

- Uses **802.1Q tagging**
 - Needs trunk port on switch
 - Each subinterface has:
 - IP address
 - VLAN ID
-

Wireless Networking Questions

3. What is the role of a Wireless LAN Controller (WLC)?

What it is

A **WLC (Wireless LAN Controller)** is a centralized device that **manages multiple wireless access points (APs)**.

Why it is used

- Simplifies wireless network management
 - Centralized control 
 - Improves security and performance
-

How it works

1. APs connect to the WLC
 2. WLC controls:
 - Configuration
 - Security policies
 - Firmware updates
 3. Traffic can be:
 - Centrally switched (via WLC)
 - Locally switched (via AP)
-

Real-life example

Think of a **school principal**  :

- Teachers (APs) handle students
 - Principal (WLC) controls policies and rules
-

Important points

- Centralized management
 - Used in large networks
 - Supports roaming (users move without disconnecting)
 - Enhances security (authentication, encryption)
-

4. What is an Access Point (AP) and how does it function?

✓ What it is

An **Access Point (AP)** is a device that provides **wireless connectivity** to devices like phones, laptops, etc.

🎯 Why it is used

- Allows wireless communication 📶
 - Connects wireless devices to wired network
-

⚙️ How it works

1. AP connects to wired network (switch/router)
 2. Broadcasts Wi-Fi signal
 3. Devices connect using SSID (network name)
 4. AP forwards data between:
 - Wireless devices ↔ wired network
-

🏠 Real-life example

Think of a **Wi-Fi router at home** 🏠:

- It sends signal
 - Your phone connects and accesses internet
-

★ Important points

- Works at **Layer 2**
- Uses radio signals 📡
- Can be:
 - Standalone AP

- Controller-based AP (with WLC)
- Supports encryption:
 - WPA2 / WPA3 

Quick Summary

Concept

Key Idea

Inter-VLAN Routing Communication between VLANs

ROAS

One router interface handles multiple VLANs

WLC

Central controller for APs

AP

Provides Wi-Fi access
